

AdamW

dense

M_t

V_t

parameter-shaped states

Adafactor

factored

reconstructed $\hat{V}_t$

$V_{c,t}$

$V_{r,t}$

row / column marginals

CAME

confidence + factored

M_t

$V_{r,t}$

$V_{c,t}$

$C_{r,t}$

$C_{c,t}$

dense M + factored V and C

APOLLO

projected

$G_t \xrightarrow{\text{projection } P_t} R_t$

M_t^R

V_t^R

projected moments $\rightarrow$ scale
on original G_t

representation $\times$ topology $\times$ update semantics